

INFORMATION FOR THE RESEARCH SEMINAR "MATHEMATICAL OPTIMISATION" (SEM 477, MAS 515)

1. GENERAL OUTLINE

Control theory is the engineering science of manipulating physical or economic systems to achieve a desired target. It has a long history in applied mathematics. While traditional approaches assumed the direct availability of a model, recent AI/ML applications advocated the consideration of online approaches.

In this seminar we will systematically study online optimisation methods arising in modern optimal control and machine learning. Opposed to offline optimization, in which the objective function is known and the goal of the decision maker is clear, the online paradigm is asking for control strategies which guarantee small regret in the long-run. This is a paradigm-shift in perceived optimal control, where optimisation algorithms are systematically used to design effective feedback controllers with long-run performance guarantees. We are reading together chapters from the books:

- Hazan, Elad, and Karan Singh. "Introduction to online nonstochastic control." arXiv preprint arXiv:2211.09619 (2022).
- Shalev-Shwartz, Shai. "Online learning and online convex optimization." Foundations and Trends in Machine Learning 4.2 (2025): 107-194.
- Rawlings, James Blake, David Q. Mayne, and Moritz Diehl. Model predictive control: theory, computation, and design. Vol. 2. Madison, WI: Nob Hill Publishing, 2020.

Depending on the background and number of students, additional material will be provided at the beginning of the semester. In particular, we encourage all students who think about writing a Master or Bachelor thesis in optimisation-related topics to attend this seminar. Please contact

Prof. Mathias Staudigl (mathias.staudigl@uni-mannheim.de) and
to express your interest and to eventually already discuss some potential topics.

2. FORMAT OF THE SEMINAR

This seminar is going to be organised as a "research seminar". We will meet on a regular basis and discuss the chapters of the book step by step together. The requirements for passing the seminar are active participation during the semester, a talk at one of the meeting slots and a seminar paper, written individually.